

Detecting Toxicity and Hazardous Contaminants in Recycled Plastics using FTIR, DSC, and Waste Water treatment

Banyan Nation

1. Introduction

The amount of plastic produced globally has increased dramatically from two million tonnes in the 1950s to over 450 million tonnes currently. A significant reason for this is due to its durability, versatility and low cost nature making it convenient to use in various industries.

The rapid growth of production has created major waste and a challenge in how it is managed. Large quantities of plastic are disposed into landfills, incinerated or released into natural environments contributing to the prevailing issue of environmental pollution and resource depletion.

To address this issue, recycling has become a critical component of sustainable economies and management. Virgin plastic is manufactured by refining crude oil and natural gas, which are extracted from fossil fuels into hydrocarbon compounds such as naphtha and then broken down into monomers through cracking. These monomers are then linked together to form polymers through a chemical process called polymerization hence creating plastic pellets. Recycling reduces the demand for virgin plastic which reduces the reliance on natural resources such as fossil fuels and lowers the amount of waste volume entering the environment. This has made plastic recycling a key element of the circular economy, a model where materials are continually recovered and reused into production systems which reduces the depletion of natural resources.

Despite the increasing rates of recycling, maintaining the quality and safety of the recycled plastic remains a challenge. These polymers originate from various waste streams and may contain additives and degradation products which can affect the performance and safety.

Plastic products contain chemical additives that improve its functionality which may remain throughout a product's lifecycle and can enter recycling streams. Common additives include plasticizers, stabilizers, adhesives, dyes and residual oils. Further, the repeated recycling process causes high exposure to heat and ultraviolet radiation which may degrade polymers, altering the material properties of the polymer. The presence of these contaminants creates an important challenge for recycling companies which is determining whether recovered plastics remain suitable for industrial use. This cannot be done through visual inspection alone, hence analytical techniques must be used to determine the properties of the polymer at a molecular and thermal level.

Banyan Nation is a plastic recycling company specialized in polyolefins. The company uses analytical testing to assess the quality, composition and safety of recycled polymers to identify contaminants which may affect the quality of the plastic

The increasing complexity of modern plastic waste streams has made analytical characterization an essential component of polymer recycling. Unlike virgin plastics, recycled polymers have often undergone prolonged environmental exposure, consumer use, and multiple processing cycles. Throughout their lifecycle, these materials may accumulate contaminants, additives, and degradation products that are not visible to the naked eye but can significantly affect their safety, quality, and performance. Consequently, recycling facilities must employ advanced analytical techniques to assess the chemical and physical properties of recycled polymers before they can be reintroduced into manufacturing processes.

Analytical tools provide scientists and engineers with the ability to examine materials at the molecular level, enabling the detection of contaminants, degradation products, and structural changes that cannot be identified through conventional inspection methods. These techniques are particularly important because recycled plastics may contain potentially hazardous substances such as plasticizers, flame retardants, heavy-metal-containing pigments, residual adhesives, and oxidation products formed during thermal processing. The presence of such compounds can compromise material performance and may pose environmental or health concerns if left undetected.

In industrial recycling facilities such as Banyan Nation, analytical characterization plays a critical role in quality assurance. By systematically evaluating recycled materials, scientists can determine whether a polymer stream has been properly sorted, whether contaminants remain after washing, and whether the polymer has undergone significant degradation during processing. This information helps ensure that recycled materials meet the specifications required for industrial applications and supports the production of high-quality recycled polymers that can effectively replace virgin plastics.

Among the most important analytical techniques used in polymer characterization are Fourier Transform Infrared Spectroscopy (FTIR), Differential Scanning Calorimetry (DSC), and Thermogravimetric Analysis (TGA). Each technique provides different but complementary information about a material. FTIR is primarily used to identify chemical composition and detect contaminants through molecular fingerprinting. DSC evaluates thermal transitions such as melting and crystallization behavior, allowing scientists to identify structural changes and contamination. TGA measures changes in mass as a material is heated, providing information about thermal stability, decomposition behavior, and the presence of additives or residual contaminants.

Together, these analytical tools provide a comprehensive understanding of recycled polymers, enabling the detection of hazardous contaminants and degradation products while ensuring material quality and consistency. Their combined use forms the foundation of modern polymer characterization and plays a crucial role in advancing safe and sustainable plastic recycling systems.

Treatment of Plastic Recycling Sludge Water Using PAC, Alum, and PAM

Introduction

During plastic recycling large volumes of sludge water are generated from the washing of the plastic bottle and chips. This sludge water contains fine particles of plastic, dirt, sand, pigments, residues and other contaminants suspended in it. Many of these particles are very small and remain throughout the water. To improve the quality of the water to ensure water recycling, coagulation and flocculation processes are used using Polyaluminium Chloride (PAC), Aluminium Sulfate (Alum) and Polyacrylamide (PAM). These chemicals work to increase the suspended particles, which are then decanted to ensure water reuse.

Chemical Background

Polyaluminium Chloride (PAC)

PAC is an aluminium based coagulant containing positively charged aluminium species that are effective at destabilizing suspended particles in wastewater. Most contaminants in the sludge water carry negative surface charge. The like charges between the particles cause them to repel each other and remain suspended in water. When PAC is added, its positive charged aluminium species neutralizes the negative charge. As the repulsion between the particles decreases they are able to move closer together and begin forming clusters known as microflocs.

Aluminium Sulfate (Alum)

Alum is an aluminium based coagulant that is used along with PAC. When it is added to water, it produces a soft gelatinous aluminium hydroxide material. The gel has a large surface area and is used to suspend the contaminants in the water. Fine plastic particles, dirt, sand and pigment are trapped in its gelatinous structure. As more particles are trapped the floc increases in size and density which allows it to settle.

Polyacrylamide (PAM)

PAM is a synthetic polymer which acts as a flocculant. It contains long molecular chains that act as a bridge for particles. As PAM disperses through the sludge water its polymer chains attach to multiple particles simultaneously causing numerous small flocs to form larger ones. The larger flocs are denser and hence settle much faster under gravity.

Experimental Procedure

The sludge water obtained from the recycling process was collected in a beaker. The sludge water was initially stirred mechanically for 15 minutes to determine how the particles would sediment. PAC, PAM and alum were then added simultaneously to the sludge water and it was stirred again for 15 minutes mechanically, ensuring contact between chemicals and contaminants. During this period the PAC coagulant neutralized particle charges, Alum formed flocs by trapping and contaminants and PAM linked the particles to form larger flocs. After mixing, the sludge water was allowed to set and slowly the flocs gradually settled to the bottom of the tank creating a sludge layer. The clearer water above the sludge was then decanted.



Sludge solution with addition of chemicals

Observations

The initial sludge water appeared to be close to opaque due to the presence of contaminants and suspended materials. Following the addition of PAC, alum and PAM flocs began to appear through the mixture.

As mixing continued the flocs became larger and denser. Compared to the untreated sludge water and sedimentation occurred more rapidly. They settled down to the bottom producing a distinct sludge layer which became visible after decantation.

Discussion

The treatment process was successful because each chemical performed a specific function.

PAC reduced the electrostatic repulsion between suspended particles, allowing them to aggregate. Alum produced a gelatinous material that trapped contaminants and increased floc formation. PAM strengthened these flocs by connecting them together through its long polymer chains.

The combined action of these chemicals increased particle size and mass, causing contaminants that would normally remain suspended to settle rapidly. This explains why sedimentation was observed to occur significantly faster in the treated sample than in untreated sludge water.

Conclusion

The use of PAC, alum, and PAM effectively improved the treatment of plastic recycling sludge water. PAC and alum destabilized and captured suspended contaminants, while PAM enhanced floc growth and settling. The treatment process resulted in faster sedimentation, reduced turbidity, and clearer water after decantation. These observations demonstrate the importance of coagulation and flocculation in industrial wastewater treatment and water recycling operations.

Detecting PP contamination in HDPE through analytical testing

FTIR Analysis of HDPE Dark Blue Samples to Determine PP Contamination

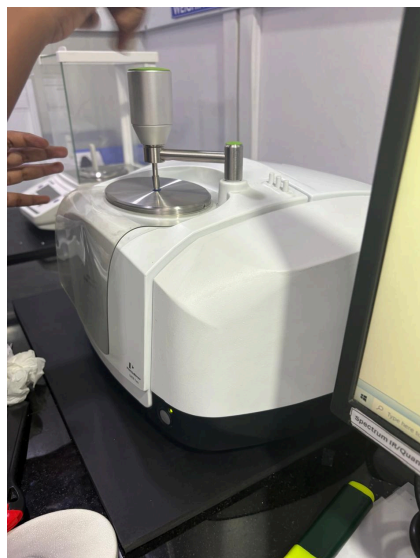
Methodology

Step 1: A handful of HDPE dark blue pellets were selected and crushed using a two roll mill.

- The sample was passed through the mill multiple times until the pellets formed a thin, flat sheet.
- This step ensured a homogeneous sample and improved the accuracy of contamination detection because individual pellets may contain varying levels of polypropylene (PP).
- The sheet was then cut into a very small square or circular piece suitable for FTIR analysis.



Two roll mill



Placement of sample in diamond crystal

Step 2: Before analysing the sample, a background scan was performed.

- The FTIR instrument first records a background spectrum to account for atmospheric interference and any contamination present in the instrument.
- This background spectrum is automatically subtracted from the sample spectrum, ensuring that only the absorbance due to the sample is measured.

Step 3: The sample was then placed on the diamond crystal of the FTIR instrument and covered.

How ATR-FTIR Works

- In ATR FTIR (Attenuated Total Reflectance Fourier Transform Infrared Spectroscopy) infrared radiation passes through a diamond crystal with a high refractive index.
- The infrared beam undergoes total internal reflection within the crystal, creating an evanescent wave that penetrates a few micrometres into the sample surface.
- Chemical bonds within the sample absorb specific infrared frequencies, producing a characteristic spectrum that can be used to identify polymers and contaminants.
- The diamond crystal is used because it is extremely hard, chemically inert, and allows consistent contact between the sample and the crystal surface.

PP contamination was identified through absorption peaks near 1350 cm^{-1} , which corresponds to radiation associated with methyl ($-\text{CH}_3$) groups present in polypropylene.

Data Table

Trials conducted 5 times

FTIR Results for L-1044 (Dark blue)							
Sample ID	Scan 1	Scan 2	Scan 3	Scan 4	Scan 5	Average	Standard deviation
L-1044 (Dark Blue)	5.89	5.83	5.72	5.73	5.82	5.80	0.07

Inference

- The five scans produced highly consistent results, with values ranging only from 5.72 to 5.89.
- The low standard deviation (0.07) indicates excellent repeatability of the FTIR measurements and suggests that the sample was relatively homogeneous after processing through the two-roll mill.
- The average contamination value of 5.80 indicates the presence of detectable polypropylene contamination within the HDPE sample.

PP%= 5.80

Differential Scanning Calorimetry (DSC) Analysis of HDPE Samples

A representative HDPE sample was weighed to obtain a mass of 18.1 mg.

- The sample was placed in an aluminium DSC pan and sealed.
- An empty aluminium pan was used as the reference.
- Nitrogen gas was continuously purged through the instrument to create an inert atmosphere and prevent oxidation during heating.



Placement of sample in DSC pan

The DSC analysis followed these seven steps:

1. Heat to 35°C.
2. Hold at 35°C for 1 minute.
3. Heat from 35°C to 220°C.
4. Hold at 220°C for 1 minute to ensure complete melting of the polymer.
5. Cool from 220°C to 35°C at a controlled rate.
6. Hold at 35°C for 1 minute.
7. Reheat from 35°C to 220°C.

The first heating cycle melts the polymer and removes any previous thermal history caused by manufacturing, processing, or storage conditions. The subsequent cooling cycle allows the polymer to recrystallise under controlled conditions. The second heating cycle is then used for analysis as it provides the thermal properties of the material without the influence of previous processing history.

Principle of DSC

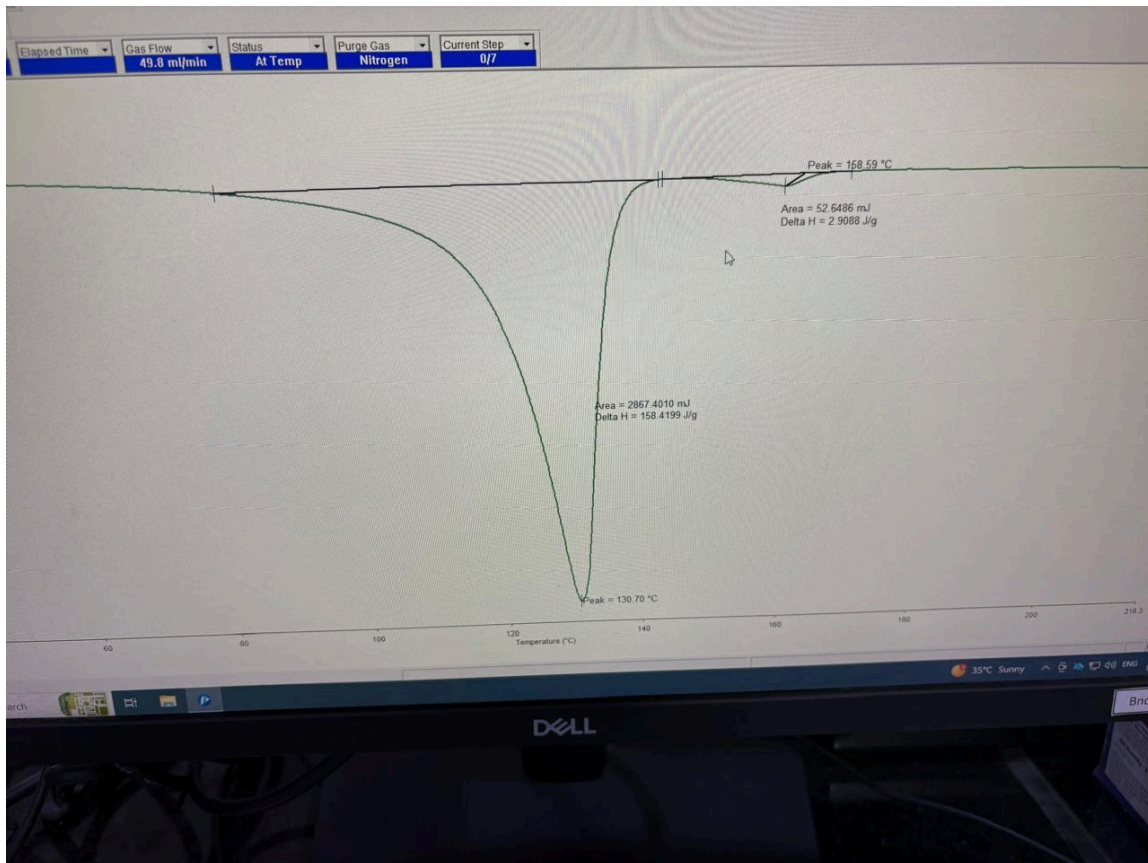
- Differential Scanning Calorimetry (DSC) measures the difference in heat flow between a sample and a reference pan as both are subjected to the same temperature programme.
- When the polymer undergoes a thermal transition, such as melting or crystallisation, it absorbs or releases heat, producing a peak on the DSC thermogram.

- The position of the melting peak corresponds to the melting temperature of the polymer, while the area of the peak represents the enthalpy.

Identification of PP Contamination

Polypropylene contamination was identified by analysing the second heating curve.

- Pure HDPE typically exhibits a melting peak at approximately 120–135°C.
- Polypropylene typically exhibits a melting peak at approximately 160–170°C.
- The presence of an additional melting peak in the polypropylene melting region indicates PP contamination within the HDPE sample.
- The size of the PP peak relative to the HDPE peak can be used to estimate the extent of contamination present in the sample.



Data Table

Peak no.	Peak type	Peak temperature (°C)	Area (mJ)	ΔH (J/g)	Polymer
Peak 1	Main melting peak	130.70	2867.4010	158.4199	HDPE
Peak 2	Secondary	168.59	52.6486	2.9088	PP

	melting peak				
--	--------------	--	--	--	--

The small peak confirms the presence of polypropylene contamination whereas the dominant peak indicates that the major part of the sample is HDPE.

The clear separation between the HDPE and PP melting peaks indicates that the polymers are present as distinct phases rather than a homogeneous blend.

Calculations

Theoretical enthalpy (ΔH) value for a 100% crystalline polymer:

HDPE: 293 J/g

PP: 209 J/g

$$\text{Weight fraction} = \frac{\Delta H_{\text{measured}}}{\Delta H_{\text{theoretical}}}$$

HDPE

$$\text{Weight of HDPE} = \frac{158.4199}{293}$$

$$\text{Weight of HDPE} = 0.5407$$

PP

$$\text{Weight of PP} = \frac{2.9088}{209}$$

$$\text{Weight of PP} = 0.0139$$

PP %

$$PP\% = \frac{W_{pp}}{W_{hdpe} + W_{pp}} \times 100$$

$$PP\% = \frac{0.0139}{0.5407 + 0.0139} \times 100$$

$$PP\% \approx 2.51\%$$

Thus, the sample contains 2-3% of PP contamination

Comparison

The FTIR measured the PP% contamination in the sample to be around 5.80% however, the testing of the sample through the DSC measured the PP% contamination to be 2.51%.

DSC measured the melting behavior, however FTIR measures chemical composition. DSC only detects PP that is crystalline in nature but FTIR detects both crystalline and amorphous (non crystalline). If polypropylene has been degraded throughout its lifecycle its melting enthalpy decreases causing DSC to underestimate contamination value. Pigments and additives in the dark blue sample may also produce a smaller DSC melting peak. FTIR analyzes the presence of methyl group characteristics regardless of how crystalline PP is. It is generally more sensitive to low contamination levels.

The two analytical techniques vary in how they measure and detect contamination one through thermal properties and one through molecular structure and hence variance in the results are expected.

Polypropylene (PP) contamination in high density polyethylene (HDPE)

If we assume the polypropylene contamination in the high density polyethylene sample to range between 2-6%, this still causes threat to the quality and effectiveness of HDPE material being used for pellets. HDPE and PP are immiscible meaning that they do not mix together at a molecular level leading to higher susceptibility to stress cracking. PP contaminants act as weak points in the HDPE matrix, making the material prone to breaking.

Compared to HDPE, PP is more brittle and hence when it is introduced into the sample it can cause breakage more easily which will result in high impact drops to cause it break. This reduces its effectiveness for being reused for bottles and can cause issues when being handled by consumers.

The melting points also vary between PP and HDPE which results in non uniform melting which creates regions of different viscosities within the melt resulting in uneven flow and inconsistent processing.

Externally, the presence of PP contamination can cause marks and bumps on the surface of the product which can reduce the product quality. The appearance of the final product may also be reduced due to streaks and flow marks because of the differing flow of the molten polymer due to different melting points.

This will cause the product to fail consumer expectations and lower its value in the market causing buyers to pay less for the product reducing profit.

